

AN **INDIAN** ADAPTATION

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Supply Chain Management

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***SUBJECT: SUPPLY CHAIN MANAGEMENT
STRATEGY***

CHAPTER: 07

CHAPTER TITLE: LOGISTICS

REFERENCE BOOK:

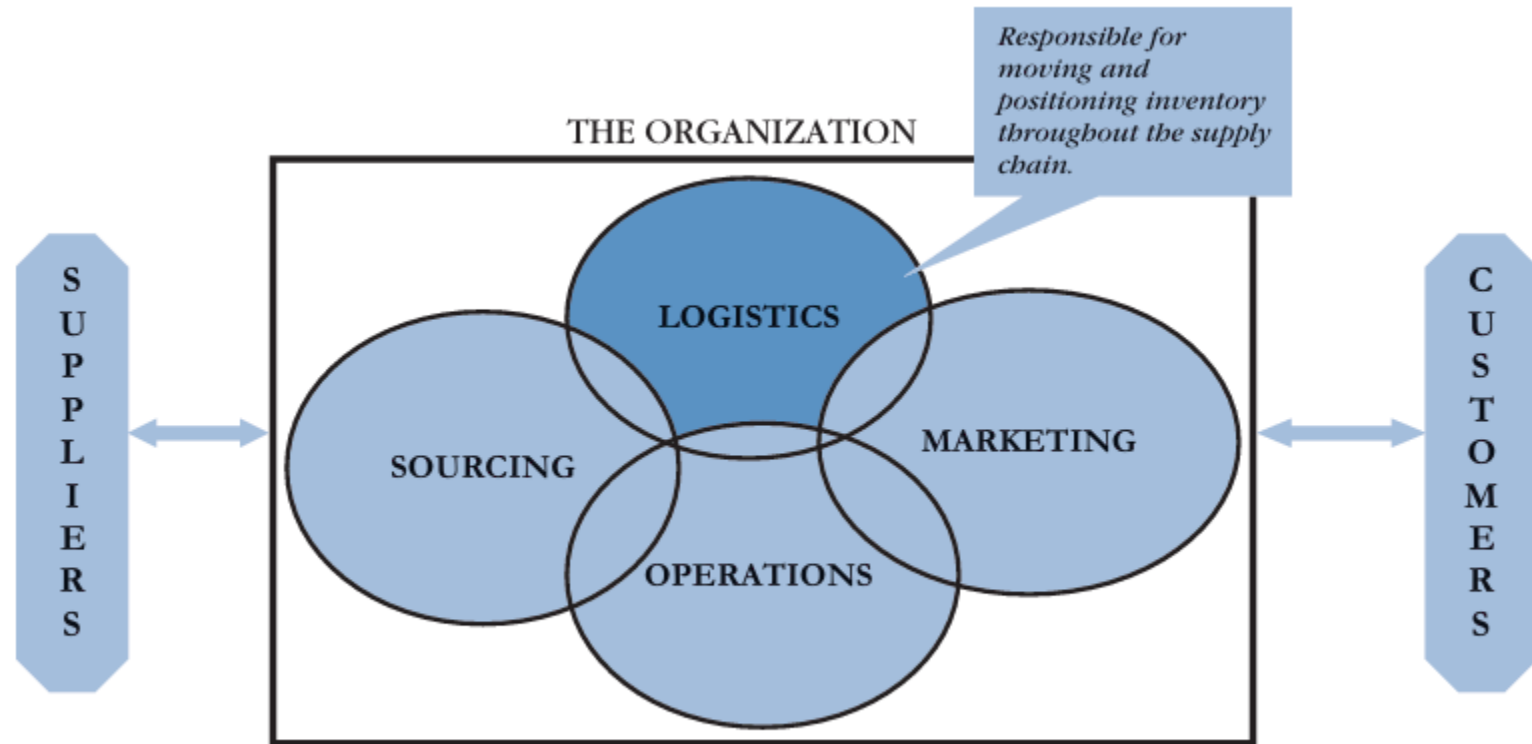
***SUPPLY CHAIN MANAGEMENT (AN INDIAN ADAPTATION), N R SANDERS & S S
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Section-A

What is Logistics

- Logistics is the business function responsible for transporting and delivering products to the right place at the right time throughout the supply chain. In essence, it is about movement and storage of product inventories throughout the chain.
- Unlike marketing that focuses on the “downstream” part of the supply chain, and sourcing that focuses on the “upstream” part, logistics connects the organization to both its customers and suppliers.
- Logistics works with sourcing to link the organization to its external suppliers and to ensure that materials are delivered inbound to the organization when needed. Logistics also works with marketing to connect the organization to the final customer by ensuring distribution and delivery of outbound products to external customers. This is shown in the Figure 1(next slide).

The Logistics Function (Figure 1)



- *The five interdependent functions of logistics* are customer response, inventory planning and management, supply, transportation, and warehousing. Figure 2 depicts each function and its objectives.

Customer Response

- Customer Response means the time and ability of a logistics system to react to customer requests or orders quickly and correctly. In other words Customer Response is the speed and efficiency with which a company fulfills customer orders and solves customer needs.
 - Customer response logistics include activities such as 1. developing and maintaining a customer service policy, 2. monitoring customer satisfaction, 3. order entry, 4. order processing, and 5. invoicing and collections.

Inventory Management

- Controlling stock levels to avoid shortages or excess inventory.
- The goal of inventory planning and management is to determine and maintain the lowest inventory level possible while meeting the customer service policy requirements.

Supply

- Supply (Procurement) – The activity of obtaining materials or products from suppliers for business operations.
- The goal of supply management is to reduce total acquisition cost while meeting the availability, response time, and quality requirements specified in the customer service policy and inventory master plan.

Transportation

- Transportation – The movement of goods from suppliers to warehouses and from warehouses to customers.
- Transportation aims to connect all pick-up and delivery to points at the lowest possible cost while adhering to customer service policy response time requirements and transportation infrastructure limitations.

Warehousing

- Storing goods safely until they are needed.
- The goal of warehousing is to reduce the cost of labor, space, and equipment in the warehouse while meeting the customer service policy's cycle time and shipping accuracy requirements as well as the inventory play's storage capacity requirements.

Logistics Functions Graph (Figure 2)



Section-B

Impact on the Organization

- Logistics has a significant impact on the organization by the mere fact that its role is to support all organizational functions by ensuring timely availability of products.
- On the inbound side of the organization logistics works with sourcing to ensure that materials needed for operations are delivered when required. On the outbound side logistics works with marketing to ensure that products are delivered to various customers when needed.

1. Impact on Operations.

- Logistics interfaces closely with operations through the planning of the amounts of inventory that must be received at various points throughout the supply chain, and the timing of those receipts.
- It must also decide on the best balance between bringing extra inventory and arranging for storage, or risking not having enough inventory.

- just-in-time (JIT) or Lean manufacturing, is known as a “pull” system where demand “pulls” the product through the supply chain. This is different from traditional systems that “push” the product through the supply chain in anticipation of demand.
- If too much inventory is delivered too soon inventory costs become high and storage becomes an issue. If too little inventory is delivered, manufacturing processes will not be able to operate without the needed materials. Therefore, it is critical that logistics understands production schedules and strikes a balance between inventory holding cost and shortages

2. Impact on Marketing.

- On the outbound side logistics provides key support for marketing and order fulfillment.
- Logistics must work with marketing to understand customer requirements, as well as storage and delivery needs.

3. Impact on Packaging.

- This has a significant impact on the ability to move and handle the material. This decision also protects the items from damage and affects both operations and marketing.

4. Impact on Finance.

- Logistics has an especially important impact on the financial picture of the organization as it is responsible for large capital expenditures, such as transportation, warehousing, and inventory.

- Its performance can be seen directly on financial measures such as return on assets (ROA) and return on investment (ROI).

Consider that ROA is computed as follows:

$$\text{ROA} = \text{Gross revenue} / \text{Assets}$$

□ From above equation we can increase ROA in three ways.

One is by increasing revenue, another by reducing expenses, and another by reducing assets. All three are directly impacted by logistics.

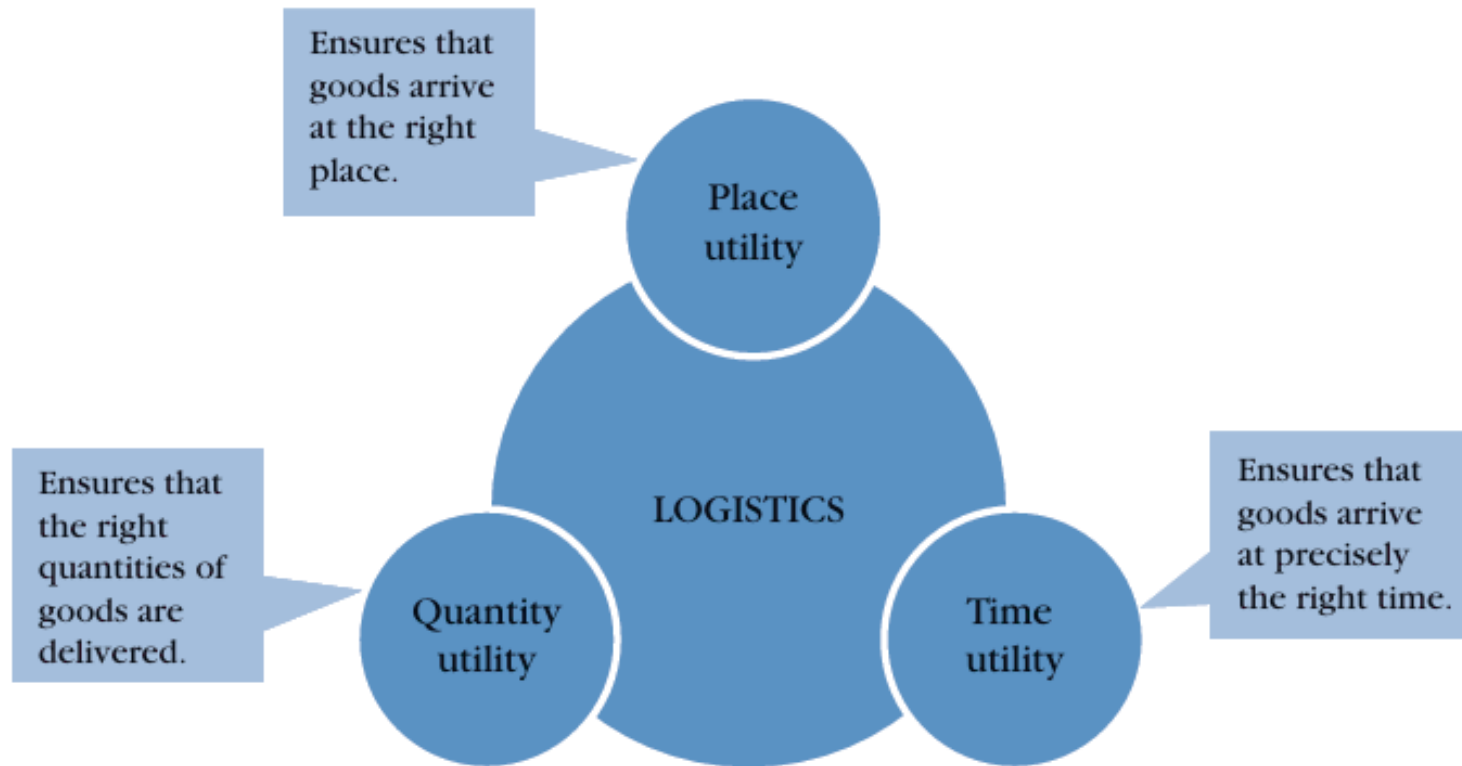
- ✓ First, by reducing inventory both expenses and assets are reduced and ROA is affected positively.
- ✓ In addition, investments in transportation and warehousing are also considered assets. By reducing these assets ROA is significantly affected. This is also another reason why some companies choose to outsource logistics.
- ✓ Finally, notice that ROA increases with an increase in revenue. Revenue can be increased through customer service and, as we already explained, logistics can play a key role in ensuring deliveries of the right quantities of goods to the right customers.

Section-C

Impact on the Supply Chain

- Logistics manages the movement of goods both downstream and upstream, and is responsible for making all the decisions that enable this to happen efficiently and effectively.
- This includes order processing and tracking, inventory management, transportation, warehousing, material handling, and packaging. Supply chain management (SCM) could not function without logistics as there would be inventory stock-outs at some locations and too much inventory at others.
- It can be said that logistics provides SCM with three utilities: place, quantity, and time, shown in Figure 3.
 - ✓ Logistics provides place utility by ensuring that goods arrive at the right place in the supply chain.
 - ✓ Similarly, logistics provides quantity utility by ensuring that the correct quantities are delivered at the right location.
 - ✓ Finally, logistics provides time utility by delivering the goods at precisely the right time needed. Recall that this is especially critical for JIT inventory systems and promotional items.

Impact of Logistics on SCM (Figure 3)

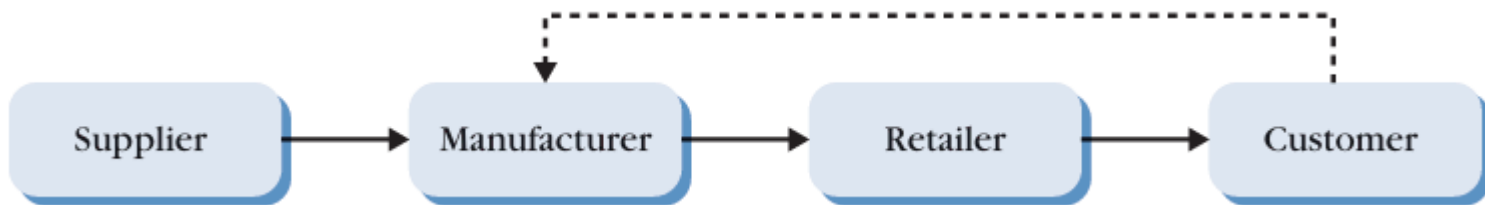


Reverse Logistics

- Reverse logistics is the process of moving products upstream from the customer back toward manufacturers and suppliers. This is the reverse direction from the way materials typically flow in a supply chain. This occurs for a variety of reasons, such as returns of damaged products or items the customer did not want.
- As with products that flow downstream, to move products upstream logistics has to organize transporting, storage, receiving, inspecting, sorting, and all other activities, to ensure efficient flow.

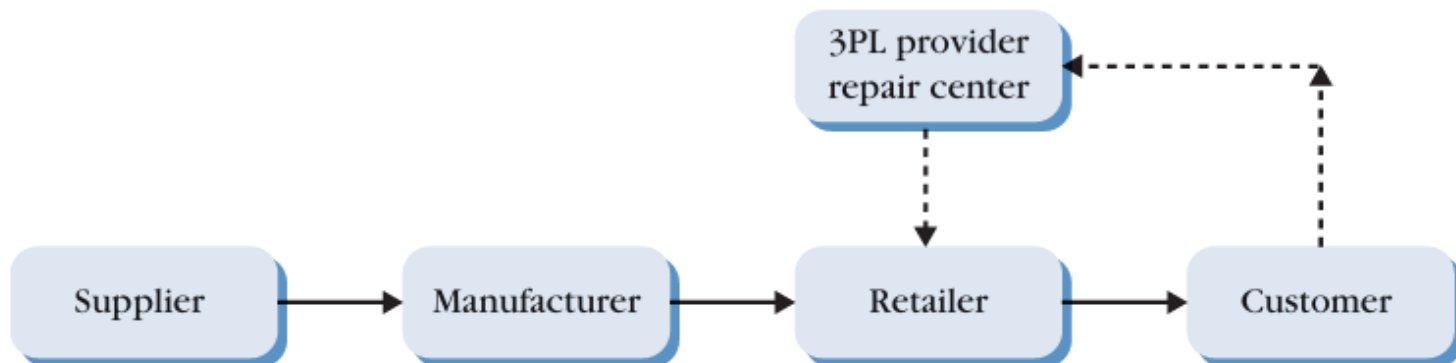
- Sometimes the items are returned directly to the manufacturer from the customer, as shown in Figure 4.

Figure 4. Reverse logistics with returns to the manufacture



- Other times a third-party-logistics (3PL) provider may be used to handle returned items and arrange for repairs bypassing the manufacturer as shown in Figure 5.

Figure 5. Reverse logistics with returns to a 3PL provider for repairs



➤ Challenges in reverse logistics

- Reverse logistics is especially challenging to design as this flow does not directly add value. Consider that there are many different types of items that are being returned to many different locations.
- There are other issues that add complexity to reverse logistics. One is handling the financials and the cash flows once items are returned.
- One more is arranging for warehouse and storage space in the reverse order that does not confuse or take away from flow of the usual downstream process.
- Another consideration is abiding by “green laws” in countries that have them that may require returning packaging materials for proper disposal.

- The role of reverse logistics has become especially important in recent years. The reason is that companies are under increasing pressure to improve customer service and have made it ever easier for customers to return goods. This is especially prevalent in retail and Internet sales, where ease of returns is a major selling feature.
- The ability to easily return goods is becoming an “order qualifier” in much of retail and it is up to logistics to continue to design ways for a cost-efficient process. For example, the success of the on-line retailer Zappos is in large part attributed to making returns convenient for customers.

Section-D

Logistics Tasks

- There are many tasks that logistics must do in order to perform the needed coordination for accurate and timely product deliveries. Here are describe the requirements of some of these tasks below.

➤ **Transportation**

- Transportation is probably the most important task logistics performs as it moves products throughout the supply chain. It is also an especially costly task given the high cost of transportation and distribution network design.
- Consider that some modes of transportation are less expensive than others, such as rail, but others are faster, such as air. The decision of which mode of transportation to use is directly tied to the consideration of the distance a product has to be moved.
- There are also other factors to consider, such as product characteristics. Perishable products, such as bananas, may need climate-controlled transportation. Highly innovative products, such as Apple's iPad, may need rapid delivery to areas with high demand. Products of very high value, such as diamonds, may need high security.

➤ **Storage**

- Directly related to transportation is the storage of goods. This involves deciding where goods will be stored, such as the number of warehouses and distribution centers, and the amounts of inventory that will be held at each center.
- We can see that the greater the number of storage areas and the wider their distribution the lower transportation costs. The key is finding an optimal balance that minimizes costs but maximizes responsiveness to customers.

➤ **Material Handling**

- Material handling is concerned with the physical handling of the product. This includes loading and unloading of goods from vehicles, placement and order picking of goods in the warehouse, and movement throughout facility, such as warehouse staging and dock areas.
- Efficient material handling saves cost, cuts down on time, and minimizes product damages in the process of movement. On the other hand, poor material handling can result in damaged products and excess processing costs.
- There is a wide range of material handling equipment, from conveyors, forklift trucks, overhead cranes, and automated storage and retrieval systems (ASRS). Unlike years ago where material handling was performed using manual labor, today many warehouses are completely automated and use very sophisticated ASRS.

➤ Packaging

- Packaging of products is an important issue that directly relates to material handling. In addition to protecting the product during transportation and storage, packaging must be compatible with material handling equipment and material handling requirements of the warehouse.
- Packaging is also directly related to modes of transportation being used. Some modes of transportation require additional packaging, such as rail or water to prevent water damage.
- Other modes, such as air, typically require less packaging. Although one mode of transportation may cost less than another, cost of packaging needs to be factored into the decision as well.
- For example, water transport is much cheaper than air, but requires packaging that may prevent water damage.

➤ **Inventory Control**

- This involves maintaining needed amounts of inventory, arranging for timely replenishments, and maintaining accurate counts of all inventories.
- Logistics is the function responsible for managing the quantities of inventory in warehouses and storage areas. Managing inventory is directly tied to transportation as arrangements must be made for timely inbound shipments of materials once stock is depleted. Logistics must also ensure accuracy of the inventory amounts.

➤ **Order Fulfilment**

- Order fulfillment is the complete process from receiving a customer order to delivering it.
- Logistics is often responsible for order fulfillment, which is completing, shipping, and delivering a customer order.
- Order fulfillment typically occurs from a warehouse or distribution center location where orders are directly placed from retailers or manufacturers.
- Logistics must ensure that the specified lead time, the time from when the order is placed until it is received, is not exceeded.

➤ **Facility Location**

- Facility location involves selecting the best site for warehouses or factories.
- Logistics is also involved in determining best location of warehouses, distribution centers, and other storage areas in relation to manufacturing facilities, customers, and suppliers.
- Decisions on location have a direct impact on the movement of products, distances traveled, cost of transportation, the ability to quickly respond to markets, and ultimately customer satisfaction.
- Location decisions also have a huge impact on transportation costs and inventory levels. It is up to logistics to determine the best locations of these facilities, considering modes of transportation to move goods between them and to the final customer.

Transportation

- Transportation is one of the most important logistics tasks as it enables logistics to provide place utility.
- Transportation deals with moving products and arranging for optimal modes of transportation.
- In developing optimal schedules, transportation must deal with both economies of scale (the larger the amount shipped at one time the lower the per unit cost) and economies of distance (the longer the distance moved at one time the lower the per unit cost).
- The challenge for logistics managers is to find an optimal balance between the economies of scale and distance, while maintaining customer service levels.

Common modes of transportation and their characteristics

➤ **Truck**

Most flexible mode

- Door-to-door delivery
- Best for short to medium distances

➤ **Water**

Used for international trade

- Low cost per unit
- Slow but economical for bulk goods

➤ **Air**

Fastest transport mode

- High cost
- Used for urgent and high-value goods

➤ **Rail**

Cost-effective for long distances

- Suitable for moving heavy loads very long distances.
- Ability to handle large and heavy items at a low cost.

➤ **Pipeline**

Used for liquids and gases

- Continuous flow
- Low operating cost after installation
- Example petroleum and natural gas.

➤ **Multimode**

- Multimode is the combined use of multiple modes of transportation for product delivery.
- A combination of a few modes can provides optimal cost and customer service.

Common modes of transportation and their characteristics

Figure 6. Cost structure by mode of transportation.

	FIXED COST	VARIABLE COST
RAIL	High <i>(e.g., equipment, terminals, tracks)</i>	Low <i>(e.g., can transport large tonnage)</i>
TRUCK	Low <i>(e.g., highways publicly supported)</i>	Medium <i>(e.g., fuel, maintenance, labor)</i>
WATER	Medium <i>(e.g., ships & equipment)</i>	Low <i>(e.g., can transport large tonnage)</i>
PIPELINE	High <i>(e.g., right-of-way, construction)</i>	Low <i>(e.g., no labor cost)</i>
AIR	High <i>(e.g., aircraft, handling, cargo)</i>	High <i>(e.g., fuel, labor, maintenance)</i>

Warehousing

➤ **Role of Warehouses in Supply Chain**

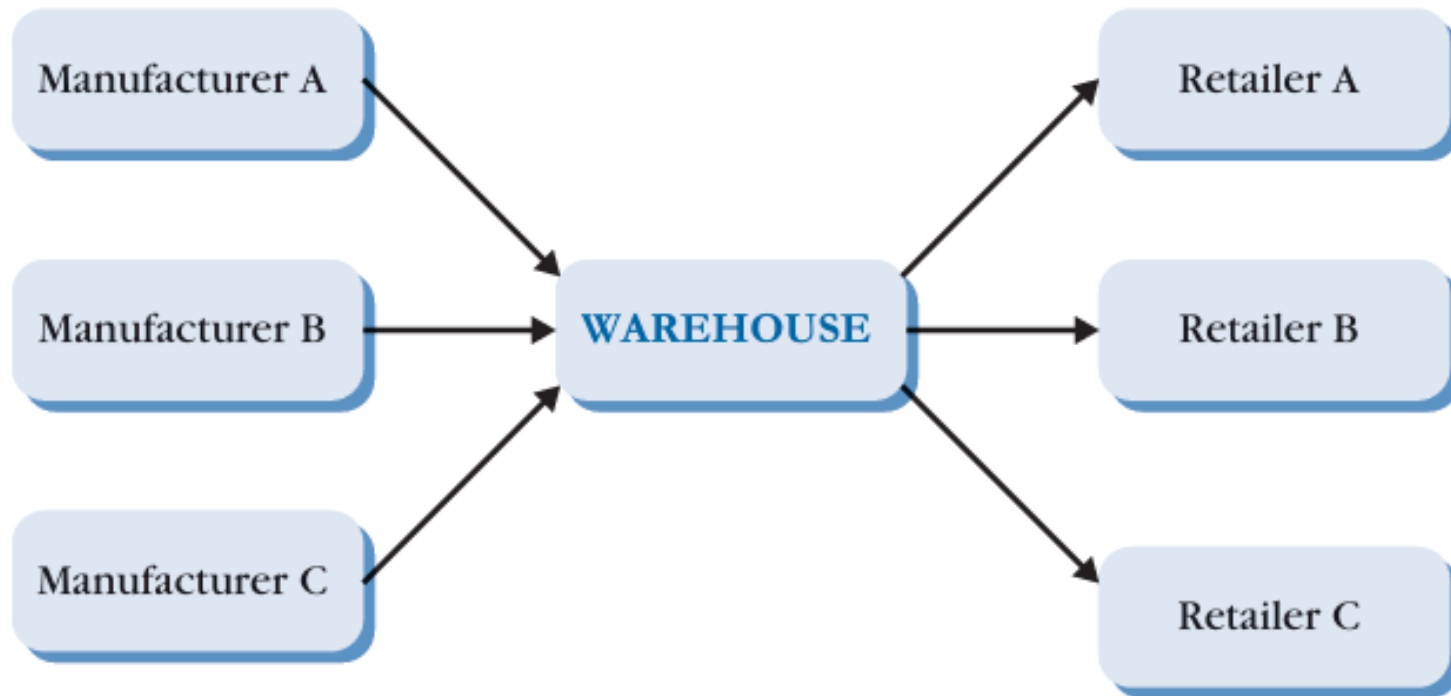
- Traditionally the role of warehouses has been to provide storage space for goods, as well as inbound and outbound transport.
- Goods that arrived at the warehouse via truck or rail were unloaded and then placed in storage bins in the warehouse. When customer orders arrived goods were picked from their storage location, known as “order picking,” then staged for transport, loaded on to trucks or rail, and shipped.
- In addition to providing storage, contemporary warehouses are increasingly places for mixing inventory assortments to meet customer needs. They are driven by enhancing and expediting movement rather than lengthy storage.
- Warehouses provide a centralized location that stores and organizes inventories of products before they get distributed to customers. For that reason they are often called distribution centers.
- Warehouses or distribution centers play a critical role in the supply chain enabling efficient movement of goods on both the inbound and outbound side of the organization.

Cross-Docking

- Two basic functions performed are breaking up of larger shipments on the inbound side, which results in lower transportation cost, and consolidation of orders for customers on the outbound side.
 - The first is called a “break-bulk” operation where a large shipment is received and then needs to be broken into smaller quantities for deliveries at multiple locations.
- Sorting of goods is yet another function performed by warehouses, enabling consolidation and mixing of orders to take place.

- One type of sorting which has gained popularity is called cross-docking.
- Cross-docking is an approach used to reconfigure bundles of product where, rather than making larger shipments, larger shipments are broken down into small shipments for local delivery in an area. This is shown in Figure 7.
- Cross docking is a logistics practice where goods are received at a warehouse or distribution center and quickly transferred directly to outbound transport, with little or no storage in between.
 - Minimal storage time
 - Reduces inventory cost
- The term implies that the products literally “cross the dock” to be loaded on to the truck for final destination and are never stored.

Figure 7. Cross Docking



Facility Location

- Location of warehouses and distribution centers are important decisions logistics has to make. An optimal location will ease distribution of goods in a timely manner, whereas a poor location will make distribution more difficult.
- There are a number of factors that must be considered.
 - ✓ The first factor to consider is proximity to customers or manufacturing facilities. It is important that the location be selected to be able to serve the largest geographic market size on the same or next day basis.
 - ✓ The cost and availability of labor is another important factor, as most warehouses and distribution centers operations are labor-intensive.
 - ✓ One of the more popular techniques is called **Factor Rating** and involves evaluating multiple alternatives based on a number of selected factors. It is a helpful procedure as it can give structure to a seemingly chaotic process when many location factors need to be considered simultaneously.

Third-party Logistics (3pl) Providers

- Third-party logistics providers, or 3PLs, are companies that provide logistics and transportation services to other firms.
- They have had an increasingly important and prominent role in the supply chain as numerous companies have outsourced their logistics function.
- External companies managing logistics
 - Provide transport, warehousing, distribution
 - Allow firms to focus on core business